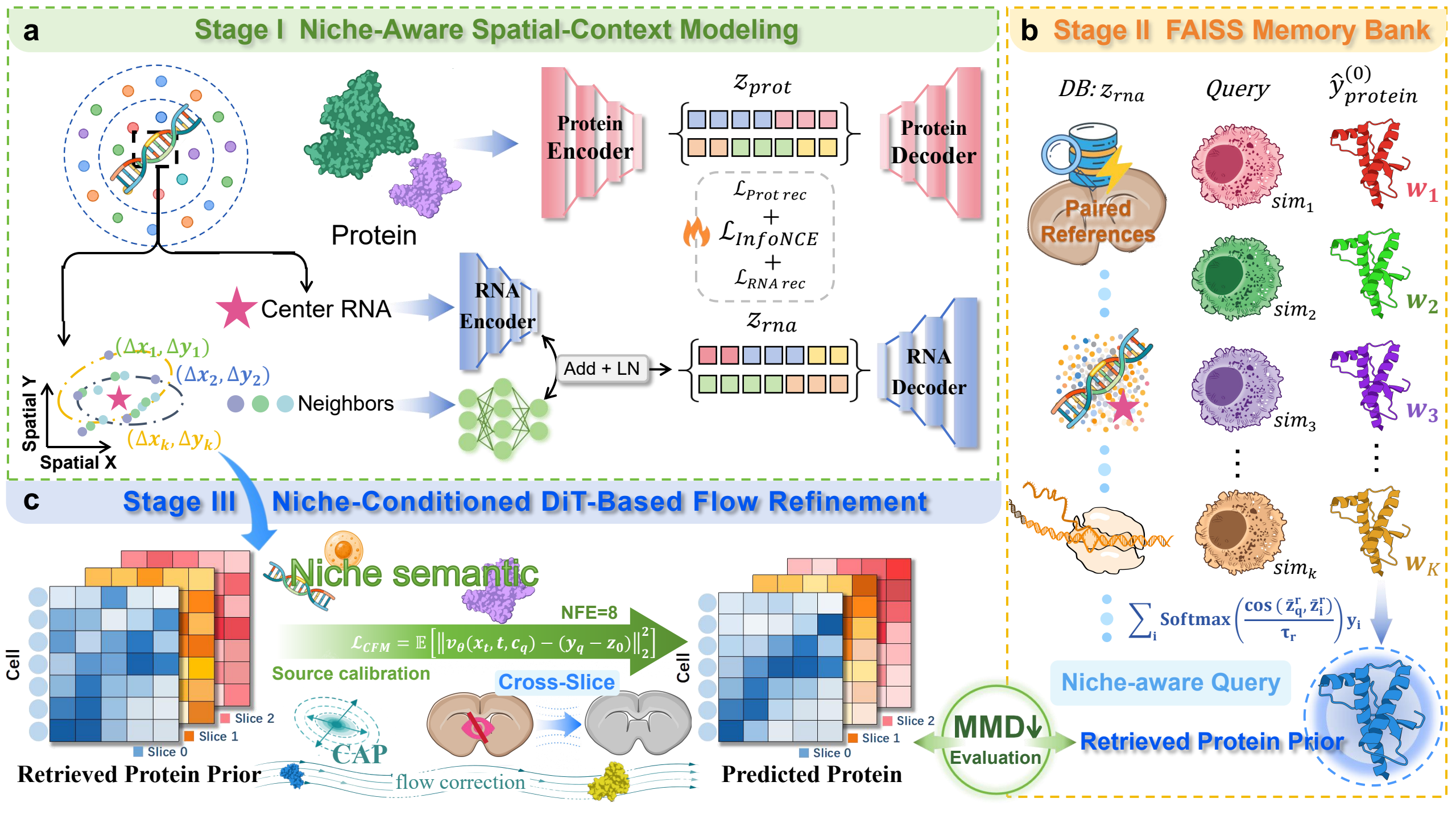

a
Stage I Niche-Aware Spatial-Context Modeling
Protein
Protein Encoder
z_{prot}
Protein Decoder
$\mathcal{L}_{Prot\,rec}$ + $\mathcal{L}_{InfoNCE}$ + $\mathcal{L}_{RNA\,rec}$
Center RNA
RNA Encoder
Add + LN
z_{rna}
RNA Decoder
$(\Delta x_1, \Delta y_1)$
$(\Delta x_2, \Delta y_2)$
$(\Delta x_k, \Delta y_k)$
Neighbors
Spatial Y
Spatial X
b
Stage II FAISS Memory Bank
DB: z_{rna}
Query
$\hat{y}^{(0)}_{protein}$
Paired References
sim_1
sim_2
sim_3
sim_k
w_1
w_2
w_3
w_K
$\sum_i \mathrm{Softmax}\left(\frac{\cos(\bar{z}^r_q, \bar{z}^r_i)}{\tau_r}\right) y_i$
Niche-aware Query
Retrieved Protein Prior
c
Stage III Niche-Conditioned DiT-Based Flow Refinement
Niche semantic
NFE=8
$\mathcal{L}_{CFM} = \mathbb{E}\left[\|v_\theta(x_t, t, c_q) - (y_q - z_0)\|_2^2\right]$
Source calibration
Cross-Slice
CAP
flow correction
Cell
Slice 2
Slice 1
Slice 0
Retrieved Protein Prior
Predicted Protein
MMD↓
Evaluation